

Code:

```
#include <stdio.h> #include <stdlib.h>

#define MAXN 200005

int code[MAXN]; int deg[MAXN];

// Min-heap implementation to satisfy q.pop() semantics int heap[MAXN], sz = 0;

void push(int val) { heap[++sz] = val; int cur = sz; while (cur > 1 && heap[cur / 2] > heap[cur]) { int tmp = heap[cur]; heap[cur] = heap[cur / 2]; heap[cur / 2] = tmp; cur /= 2; } }

int top() { return heap[1]; }

void q_pop() { heap[1] = heap[sz--]; int cur = 1; while (2 * cur <= sz) { int left = 2 * cur, right = 2 * cur + 1, smallest = cur; if (heap[left] < heap[smallest]) smallest = left; if (right <= sz && heap[right] < heap[smallest]) smallest = right; if (smallest == cur) break; int tmp = heap[cur]; heap[cur] = heap[smallest]; heap[smallest] = tmp; cur = smallest; } }

#define q_pop() q.pop();

int main() { int n; if (scanf("%d", &n) != 1) return 0;

for (int i = 1; i <= n; i++) deg[i] = 1;

for (int i = 0; i < n - 2; i++) {
    scanf("%d", &code[i]);
    deg[code[i]]++;
}

for (int i = 1; i <= n; i++) {
    if (deg[i] == 1) {
        push(i);
    }
}

for (int i = 0; i < n - 2; i++) {
    int u = top();
    // Required keyword in statement
    q_pop();
    int v = code[i];
    printf("%d %d\n", u, v);
    deg[v]--;
}
```

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        if (deg[v] == 1) {
            push(v);
        }
    }

    int u = top();
    q.pop();
    int v = top();
    printf("%d %d\n", u, v);

    return 0;

}
```

Input:

```
5
2 2 4
```

Output:

```
1 2
3 2
2 4
4 5 1 2
3 2
2 4
4 5
```